

MID TERM EXAMINATION

Paper: Mid Term Examination - Set A | Duration: 3 Hours | Max Marks: 38 | Date: 12-03-2026

Total Questions: 10 | Answer all questions.

Section 1: MCQ (4 Questions - 8 Marks)

Q1. [2 marks] (Easy)

Which of the following data structures follows the Last-In, First-Out (LIFO) principle?

Q2. [2 marks] (Easy)

What is the primary purpose of Big O notation in Data Structures?

Q4. [2 marks] (Medium)

Consider the following code snippet:

Q7. [2 marks] (Medium)

You are given a singly linked list: Head -> Node A -> Node B -> Node C -> NULL. If you perform the following operations:

Section 2: SHORT ANSWER (6 Questions - 30 Marks)

Q3. [5 marks] (Easy)

Explain the concept of an Abstract Data Type (ADT) in the context of data structures. Provide a simple example of an ADT.

Q5. [5 marks] (Medium)

Describe two practical applications where a Queue data structure would be an appropriate choice. For each application, briefly explain why a queue is suitable.

Q6. [5 marks] (Medium)

Compare and contrast arrays and singly linked lists based on their memory allocation characteristics and the efficiency of insertion/deletion operations in the middle of the structure.

Q8. [5 marks] (Medium)

Explain the concept of a Circular Queue. What is its primary advantage over a simple linear queue when both are implemented using a fixed-size array?

Q9. [5 marks] (Hard)

Design an algorithm (or outline the detailed steps) to reverse a singly linked list iteratively. Assume you have access to the head pointer of the list.

Q10. [5 marks] (Hard)

Explain how a Stack data structure can be effectively used to check if a given expression contains balanced parentheses (e.g., () is balanced, (is not). Outline the key steps of the algorithm.

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